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Printed circuit board and electronic device including the same

Abstract

An electronic device is provided. The electronic device includes an electronic component including a coil therein, and a printed circuit board that is disposed to surround at least a part of the electronic component and includes plural slits disposed at preset intervals in at least one of regions facing the electronic component, wherein each of the plural slits may include a central slit portion extended along a center of the slit, and a symmetric slit portion extended to be symmetric with respect to the central slit portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2023/010117, filed on Jul. 14, 2023, which is based on and claims the benefit of a Korean patent application number filed on Jul. 20, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0107038, filed on Aug. 25, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The disclosure relates to a printed circuit board and an electronic device including the same.

BACKGROUND ART

(2) A portable electronic device including a speaker (e.g., smartphone) may be designed in a structure where a portion of a printed circuit board included in the portable electronic device is cut off and a speaker is placed thereat in order to make the thickness thin. For example, the cut portion of the printed circuit board may be disposed to surround at least a portion of the circumference of the speaker.

(3) The portable electronic device may include a battery for supplying power to a power consuming component (e.g., power amplifier) disposed on the printed circuit board. As power is supplied from the battery to the power consuming component, a current path from the battery to the power consuming component and a return current path from the power consuming component to the ground portion of the printed circuit board may be formed.

(4) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

DISCLOSURE

Technical Problem

(5) A part of the return current path spreading to the ground portion of the printed circuit board may be formed at the cut portion of the printed circuit board for arranging the speaker. A return current may flow through the portion of the printed circuit board being cut off (e.g., a portion of the printed circuit board surrounding the speaker), and thus an H-field may be formed due to the return current. The H-field passes through a coil inside the speaker and generates a current in the coil, which may cause unintended speaker operation.

(6) The H-field applied to the speaker may be proportional to the intensity of current and inversely proportional to the square of the distance (e.g., distance between the speaker and the return current path). Since the gap between the printed circuit board and the speaker is minimized by design so as to make the portable electronic device thin and light, it may be difficult to use a method of increasing the distance between the speaker and the return current path to attenuate the H-field. Therefore, there is a need for a structure capable of reducing the intensity of a return current

generated on the printed circuit board in order to attenuate the H-field.

(7) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device for a structure capable of attenuating an H-field caused by a return current by blocking current flows around the speaker.

(8) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Technical Solution

(9) In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes an electronic component including a coil therein, and a printed circuit board that is disposed to surround at least a part of the electronic component and includes plural slits disposed at preset intervals in at least one of regions facing the electronic component, wherein each of the plural slits may include a central slit portion extended along a center of the slit, and a symmetric slit portion extended to be symmetric with respect to the central slit portion.

(10) In accordance with another aspect of the disclosure, a printed circuit board is provided. The printed circuit includes a cut region formed by cutting off a portion of the printed circuit board, and a plurality of slits formed around the cut region, wherein each of the plural slits may include a central slit portion extended along a center of the slit and a symmetric slit portion extended to be symmetric with respect to the central slit portion.

Advantageous Effects

(11) An electronic device according to an embodiment of the disclosure may include a printed circuit board having slits formed thereon to thereby attenuate H-fields formed in electronic components (e.g., speaker, receiver).

(12) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

DESCRIPTION OF DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;

(3) FIG. 2 is a view illustrating an electronic device according to an embodiment of the disclosure;

(4) FIGS. 3A, 3B, 3C and 3D are views illustrating an electronic component and a printed circuit board according to various embodiments of the disclosure;

(5) FIG. 4 is a view showing slits and protrusions according to an embodiment of the disclosure;

(6) FIG. 5 is a view illustrating a printed circuit board including slits according to an embodiment of the disclosure;

(7) FIG. 6 is a view illustrating a printed circuit board including a first slit and a second slit according to an embodiment of the disclosure;

(8) FIG. 7 is a view illustrating a protrusion including a circular protrusion portion according to an embodiment of the disclosure;

(9) FIGS. 8A, 8B, 8C and 8D are diagrams illustrating a return current generated on a printed circuit board and an equivalent circuit according to various embodiments of the disclosure; and

(10) FIGS. 9A and 9B are diagrams illustrating the intensity of the H-field generated on a printed circuit board according to various embodiments of the disclosure.

(11) The same reference numerals are used to represent the same elements throughout the drawings.

MODE FOR DISCLOSURE

(12) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the present disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

(13) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(14) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

(15) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

(16) Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an external electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an external electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment of the disclosure, the electronic device **101** may communicate with the external electronic device **104** via the server **108**. According to an embodiment of the disclosure, the electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments of the disclosure, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments of the disclosure, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(17) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment of the disclosure, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an embodiment of the disclosure, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary

processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(18) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., a sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(19) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(20) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(21) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(22) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment of the disclosure, the receiver may be implemented as separate from, or as part of the speaker.

(23) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment of the disclosure, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

(24) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment of the disclosure, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(25) The sensor module **176** may detect an operational state (e.g., power or temperature) of the

electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment of the disclosure, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(26) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment of the disclosure, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(27) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the external electronic device **102**). According to an embodiment of the disclosure, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(28) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment of the disclosure, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(29) The camera module **180** may capture a still image or moving images. According to an embodiment of the disclosure, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(30) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment of the disclosure, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(31) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment of the disclosure, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(32) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment of the disclosure, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5th generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using

subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(33) The wireless communication module **192** may support a 5G network, after a 4th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter (mm) Wave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the external electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment of the disclosure, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(34) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment of the disclosure, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment of the disclosure, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment of the disclosure, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(35) According to an embodiment of the disclosure, the antenna module **197** may form a mmWave antenna module. According to an embodiment of the disclosure, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(36) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(37) According to an embodiment of the disclosure, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment of the disclosure, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a

request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment of the disclosure, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment of the disclosure, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(38) FIG. **2** is a view illustrating an electronic device **200** according to an embodiment of the disclosure.

(39) In an embodiment of the disclosure, the electronic device **200** of FIG. **2** may refer to the electronic device **101** of FIG. **1** or may be an electronic device including at least some of the electronic device **101** of FIG. **1**.

(40) In describing the electronic device **200** according to an embodiment of the disclosure, the width direction of the electronic device **200** may mean the x-axis direction, and the length direction of the electronic device **200** may mean the y-axis direction.

(41) In an embodiment of the disclosure, the electronic device **200** may include a battery **210**, a camera module **220**, an electronic component **230**, a printed circuit board **240**, a shield can **250**, an antenna connection part **260**, and/or a battery connector **270**.

(42) In an embodiment of the disclosure, the battery **210** in FIG. **2** may indicate the battery **189** in FIG. **1**. The battery **210** may supply power to at least one component of the electronic device **200**. For example, the battery **210** may supply power to a power consuming component disposed on the printed circuit board **240**.

(43) In an embodiment of the disclosure, the camera module **220** may capture still images and moving images. The camera module **220** in FIG. **2** may mean the camera module **180** in FIG. **1** or may include at least some of the camera module **180** in FIG. **1**. For example, the camera module **220** may include one or more lenses, image sensors, image signal processors, and/or flashes.

(44) In an embodiment of the disclosure, the electronic component **230** may be a component including a coil therein. For example, the electronic component **230** may be an acoustic instrument including a coil (e.g., speaker, receiver).

(45) In an embodiment of the disclosure, the shield can **250** may be formed in a structure surrounding the exterior of a component disposed on the printed circuit board **240**. For example, to shield noise generated from a power consuming component disposed on the printed circuit board **240**, the shield can **250** may be disposed to surround the exterior of the power consuming component. Multiple power consuming components may be disposed inside the shield can **250**, and the shield can **250** may serve to shield noise generated from the multiple power consuming components. The power consuming component may be a component that consumes relatively more power than other components of the electronic device **200**. For example, the power consuming component may be a power amplifier (PA).

(46) In an embodiment of the disclosure, the printed circuit board **240** may be disposed to surround at least a portion of the electronic component **230**. For example, the printed circuit board **240** may surround the outer periphery of the electronic component **230** by being spaced apart from the outer

circumference of the electronic component **230** in the x-axis direction and/or the y-axis direction.

(47) In an embodiment of the disclosure, one direction of the printed circuit board **240** may indicate a negative x-axis direction with respect to the printed circuit board **240**, and the other direction of the printed circuit board **240** may indicate a positive x-axis direction with respect to the printed circuit board **240**.

(48) In an embodiment of the disclosure, the camera module **220** may be positioned in one direction of the printed circuit board **240**. For example, the camera module **220** may be positioned in a negative x-axis direction with respect to the printed circuit board **240**.

(49) In an embodiment of the disclosure, the antenna connection part **260** may be extended toward a power consuming component positioned inside the shield can **250** in the other direction (e.g., positive x-axis direction) of the printed circuit board **240**, and may be disposed on one surface of the printed circuit board **240**. The antenna connection part **260** may serve to electrically connect an antenna (e.g., antenna module **197** in FIG. 1) and the printed circuit board **240**.

(50) With reference to FIG. 2, the battery **210** may be disposed to be spaced apart in a longitudinal direction of the electronic device **200** with respect to the camera module **220**, the electronic component **230**, and/or the printed circuit board **240**. For example, the battery **210** may be disposed in a negative y-axis direction with respect to the camera module **220**, the electronic component **230**, and/or the printed circuit board **240**.

(51) In various embodiments, the battery connector **270** may be disposed in one direction of the battery **210**. For example, in FIG. 2, the battery connector **270** may be positioned in a positive y-axis direction with respect to the battery **210**.

(52) In an embodiment of the disclosure, the battery connector **270** may be connected to the battery **210** at one end thereof and connected to the printed circuit board **240** at the other end thereof. The battery connector **270** may serve to supply power generated from the battery **210** to the printed circuit board **240** by electrically connecting the battery **210** and the printed circuit board **240**.

(53) Although the battery **210** is shown as being positioned in a negative y-axis direction with respect to the printed circuit board **240** in FIG. 2, the relative positional relationship between the printed circuit board **240** and the battery **210** is not limited thereto. For example, the battery **210** may be positioned in a positive y-axis direction with respect to the printed circuit board **240**.

(54) FIGS. 3A, 3B, 3C and 3D are views illustrating an electronic component **230** and a printed circuit board **240** according to various embodiments of the disclosure.

(55) In describing the printed circuit board **240** according to an embodiment of the disclosure, the width direction of the printed circuit board **240** may refer to the x-axis direction, and the length direction of the printed circuit board **240** may refer to the y-axis direction.

(56) In an embodiment of the disclosure, the printed circuit board **240** may include a cut region **241** at least in part. The cut region **241** may be a region formed by cutting off a portion of the printed circuit board **240**. For example, in FIG. 3A, a portion of the printed circuit board **240** may be cut off in the longitudinal and width directions of the printed circuit board **240** to form a cut region **241**. Referring to FIG. 3A, the cut region **241** is illustrated as being formed in a rectangular shape, but this is illustrative and the shape of the cut region **241** may be not limited thereto.

(57) In an embodiment of the disclosure, the electronic component **230** may be disposed in the cut region **241** of the printed circuit board **240**. The electronic component **230** may be disposed in the cut region **241** and surrounded by the printed circuit board **240**. In the cut region **241**, the electronic component **230** may be positioned to be spaced apart from the printed circuit board **240**.

(58) In an embodiment of the disclosure, a first direction of the electronic component **230** may indicate a negative x-axis direction with respect to the electronic component **230**. A second direction of the electronic component **230** may indicate a positive x-axis direction with respect to the electronic component **230**. A third direction of the electronic component **230** may indicate a negative y-axis direction with respect to the electronic component **230**. A fourth direction of the electronic component **230** may indicate a positive y-axis direction with respect to the electronic

component **230**.

(59) In an embodiment of the disclosure, a first direction of the cut region **241** may refer to a negative x-axis direction with respect to the cut region **241**, and a second direction of the cut region **241** may refer to a positive x-axis direction with respect to the cut region **241**. A third direction of the cut region **241** may indicate a negative y-axis direction with respect to the cut region **241**. A fourth direction of the cut region **241** may indicate a positive y-axis direction with respect to the cut region **241**.

(60) In an embodiment of the disclosure, the printed circuit board **240** may include a first region **241a**, a second region **241b**, and/or a third region **241c**, which are regions surrounding the outer periphery of the electronic component **230**.

(61) In an embodiment of the disclosure, the first region **241a** may refer to a region of the printed circuit board **240** that is spaced apart in the first direction of the electronic component **230** to face the electronic component **230**. The second region **241b** may refer to a region of the printed circuit board **240** that is spaced apart in the second direction of the electronic component **230** to face the electronic component **230**. The third region **241c** may refer to a region of the printed circuit board **240** that is spaced apart in the third direction of the electronic component **230** to face the electronic component **230**.

(62) In an embodiment of the disclosure, the printed circuit board **240** may include slits **242** at least in part. For example, plural slits **242** may be arranged at intervals in at least one of the first region **241a**, the second region **241b**, or the third region **241c** of the printed circuit board **240**.

(63) In an embodiment of the disclosure, the printed circuit board **240** may include a ground at least in part. Alternatively, a ground, as a separate element from the printed circuit board **240**, may be positioned around the outer periphery of the printed circuit board **240**. The ground may contain a metallic material (e.g., copper) therein.

(64) In an embodiment of the disclosure, the slit **242** may be a T-shaped groove formed by cutting off at least a portion of the ground. A non-conductive material may be disposed inside the slit **242** (e.g., space of slit groove).

(65) FIGS. **3A**, **3B**, **3C** and **3D** depict that a portion of the printed circuit board **240** is directly cut off to form a slit **242**, but this may be illustrative for explaining the position of the slit **242**. For example, the slit **242** may be formed, not by directly cutting off a portion of the printed circuit board **240**, but by cutting off a portion of the ground located around the outer periphery of the printed circuit board **240**.

(66) Referring to FIG. **3A**, in an embodiment of the disclosure, a plurality of slits **242** may be formed in the first region **241a** of the printed circuit board **240**. For example, plural slits **242** may be disposed at specified intervals in the first region **241a** along the length direction (e.g., y-axis direction) of the printed circuit board **240**. Although the printed circuit board **240** is shown as including four slits **242** in FIG. **3A**, this is illustrative and the number of slits **242** included in the printed circuit board **240** may be not limited thereto.

(67) Referring to FIG. **3B**, in an embodiment of the disclosure, a plurality of slits **242** may be formed in the third region **241c** of the printed circuit board **240**. For example, plural slits **242** may be disposed at specified intervals in the third region **241c** along the width direction (e.g., x-axis direction) of the printed circuit board **240**. Although the printed circuit board **240** is shown as including eight slits **242** in FIG. **3B**, this is illustrative and the number of slits **242** included in the printed circuit board **240** may be not limited thereto.

(68) Referring to FIG. **3C**, in an embodiment of the disclosure, a plurality of slits **242** may be formed in the second region **241b** of the printed circuit board **240**. For example, plural slits **242** may be disposed at specified intervals in the second region **241b** along the length direction (e.g., y-axis direction) of the printed circuit board **240**. Although the printed circuit board **240** is shown as including four slits **242** in FIG. **3C**, this is illustrative and the number of slits **242** included in the printed circuit board **240** may be not limited thereto.

(69) Referring to FIG. 3D, in an embodiment of the disclosure, a plurality of slits **242** may be formed in each of the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**. For example, plural slits **242** may be disposed at specified intervals in the first region **241a**, the second region **241b**, and the third region **241c** along the width direction (e.g., x-axis direction) or the length direction (e.g., y-axis direction) of the printed circuit board **240**.

(70) In an embodiment of the disclosure, among the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**, plural slits **242** may be formed in a region located relatively close to a component consuming a lot of power (e.g., power consuming component positioned inside the shield can **250** in FIG. 2). For example, when the first region **241a** is located closer to the power consuming component than other regions (e.g., second region **241b**, third region **241c**), a plurality of slits **242** may be formed in the first region **241a**.

(71) FIGS. 3A, 3B, 3C and 3D depict that slits **242** are disposed in the first direction, the second direction, and/or the third direction with respect to the electronic component **230**, but the position where slits **242** are disposed on the printed circuit board **240** may be not limited thereto. For example, slits **242** may be formed in a region of the printed circuit board **240** positioned in the fourth direction (e.g., positive y-axis direction) with respect to the electronic component **230**.

(72) FIG. 4 is a view showing slits **242** and protrusions **243** according to an embodiment of the disclosure.

(73) In an embodiment of the disclosure, slits **242** and protrusions **243** may be formed in each of the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**.

(74) Portion 'A' in FIG. 4 indicates an enlarged view of a slit **242** and a protrusion **243** formed in the third region **241c** of the printed circuit board **240**.

(75) In an embodiment of the disclosure, a slit **242** may include a central slit portion **2421** and/or a symmetric slit portion **2422**. The slit **242** may refer to an opening or gap formed at a portion of the printed circuit board **240** and having a pre-specified length.

(76) In an embodiment of the disclosure, the central slit portion **2421** may refer to a portion formed to extend substantially perpendicular to one surface of the printed circuit board **240**. For example, when a part of the printed circuit board **240** is extended in the x-axis direction, the central slit portion **2421** may be formed to extend in the y-axis direction perpendicular to the x-axis direction. Referring to FIG. 4, the central slit portion **2421** of the slit **242** positioned in the third direction (e.g., negative y-axis direction) of the electronic component **230** may be formed to extend in a direction perpendicular to a part of the printed circuit board **240** (e.g., y-axis direction).

(77) In an embodiment of the disclosure, the central slit portion **2421** of the slit **242** may refer to a portion formed to extend in a direction away from the electronic component **230**. For example, when the electronic component **230** is positioned in the positive y-axis direction with respect to the slit **242**, the central slit portion **2421** may be formed to extend toward the negative y-axis direction.

(78) In an embodiment of the disclosure, the central slit portion **2421** may be a portion extended along the center of the slit **242**. For example, for a slit **242** positioned in the first region **241a** or the second region **241b**, the center of the slit **242** indicates the center in the y-axis direction, and the central slit portion **2421** may be formed to extend along the x-axis direction with respect to the center of the slit **242** in the y-axis direction. For a slit **242** positioned in the third region **241c**, the center of the slit **242** indicates the center in the x-axis direction, and the central slit portion **2421** may be formed to extend along the y-axis direction with respect to the center of the slit **242** in the x-axis direction.

(79) In an embodiment of the disclosure, the symmetric slit portion **2422** may be formed to extend from an end of the central slit portion **2421** perpendicularly to the direction in which the central slit portion **2421** is extended in a form symmetric about the central slit portion **2421**. For example, the symmetric slit portion **2422** of a slit **242** positioned in the third direction (e.g., negative y-axis direction) of the electronic component **230** may be formed to extend, from an end of the central slit

portion **2421** toward the negative y-axis direction, in a direction perpendicular to the central slit portion **2421** (e.g., x-axis direction).

(80) In an embodiment of the disclosure, a protrusion **243** may be formed between plural slits **242**. The protrusion **243** may be formed by a portion of the printed circuit board **240** that protrudes and is extended in a direction toward the electronic component **230**. The protrusion **243** may include a central protrusion portion **2431** and/or a symmetric protrusion portion **2432** at least in part.

(81) In an embodiment of the disclosure, the protrusion **243** may be a portion of the printed circuit board **240** that protrudes and is extended around a slit **242** when the slit **242** is formed at a portion of the printed circuit board **240**. For example, the central protrusion portion **2431** may be a portion formed by symmetric slit portions **2422**. The symmetric protrusion portion **2432** may be a portion formed by central slit portions **2421**. Since the symmetric slit portion **2422** is extended further toward the protrusion **243** compared to the central slit portion **2421**, the width (e.g., length in x-axis direction) of the central protrusion portion **2431** may be formed to be smaller than the width (e.g., length in x-axis direction) of the symmetric protrusion portion **2432**.

(82) In an embodiment of the disclosure, slits **242** may each be positioned in one direction of a protrusion **243** (e.g., negative x-axis direction relative to the protrusion **243**) and in the other direction (e.g., positive x-axis direction relative to the protrusion **243**). For example, symmetric slit portions **2422** may each be positioned in one direction of a central protrusion portion **2431** and in the other direction. Central slit portions **2421** may each be positioned in one direction of a symmetric protrusion portion **2432** and in the other direction.

(83) In an embodiment of the disclosure, the central protrusion portion **2431** may be formed by a portion of the printed circuit board **240** that protrudes and is extended in a direction substantially perpendicular to the direction in which the printed circuit board **240** is extended. For example, as indicated by portion 'A' in FIG. 4, the central protrusion portion **2431** may be formed by a portion of the printed circuit board **240** that protrudes and is extended in a direction (e.g., y-axis direction) substantially perpendicular to the direction (e.g., x-axis direction) in which the printed circuit board **240** is extended.

(84) In an embodiment of the disclosure, the symmetric protrusion portion **2432** may be formed to extend in a direction substantially perpendicular to the direction in which the central protrusion portion **2431** is extended, in a form symmetric with respect to the central protrusion portion **2431**. For example, the symmetric protrusion portion **2432** may be formed to extend in the negative x-axis direction and the positive x-axis direction from an end of the central protrusion portion **2431** (e.g., the end positioned in the positive y-axis direction of the central protrusion portion **2431**) so as to be symmetric about the central protrusion portion **2431**.

(85) In an embodiment of the disclosure, the symmetric protrusion portion **2432** may have a symmetrical shape with respect to the central protrusion portion **2431**. For example, in the symmetric protrusion portion **2432**, the length extended in the negative x-axis direction from the central protrusion portion **2431** and the length extended in the positive x-axis direction may be equal to each other.

(86) Referring to FIG. 4, the symmetric protrusion portion **2432** is illustrated as extending in a rectangular shape, but this is illustrative, and the shape of the symmetric protrusion portion **2432** may be not limited thereto. For example, the symmetric protrusion portion **2432** may be formed to be symmetric with respect to the center of the protrusion **243** in the x-axis direction so as to have one of various shapes (e.g., ellipse, polygon).

(87) In an embodiment of the disclosure, one end of the protrusion **243** may indicate the end of the protrusion **243** positioned closest to the cut region **241**. For example, one end of the protrusion **243** in portion 'A' may indicate the end toward the positive y-axis direction.

(88) In an embodiment of the disclosure, a current may flow in the protrusion **243** of the printed circuit board **240**. For example, a first current C1 may flow toward one end of the protrusion **243**. The first current C1 may flow toward one end of the protrusion **243** along the positive y-axis

direction.

(89) In an embodiment of the disclosure, the first current **C1** as a return current (C-r, see FIG. **8A**) may be a current having a frequency lower than a preset reference value. Referring to FIG. **4**, the first current **C1** may include a current component flowing in parallel to the y-axis direction and current components flowing in the positive x-axis direction and the negative x-axis direction. The length by which the protrusion **243** is extended (e.g., length extended in y-axis direction in portion 'A') may be formed to a preset length or less, which can prevent formation of an H-field caused by a current component of the first current **C1** flowing in parallel to the y-axis direction. Among the first current **C1**, current components flowing in the positive x-axis direction and the negative x-axis direction have vectors symmetric about the x-axis center and the same scalar so they can cancel each other out.

(90) In an embodiment of the disclosure, one direction of a protrusion **243** may indicate the negative x-axis direction with respect to the protrusion **243**, and the other direction of a protrusion **243** may indicate the positive x-axis direction with respect to the protrusion **243**.

(91) Referring to FIG. **4**, a second current **C2** may flow in a direction where the protrusion **243** is extended (e.g., positive y-axis direction) and then flow in a direction (e.g., x-axis direction) perpendicular to the direction where the protrusion **243** is extended (e.g., positive y-axis direction). For example, the second current **C2** may flow along a direction where the symmetric protrusion portion **2432** is extended (e.g., negative x-axis direction or positive x-axis direction).

(92) In an embodiment of the disclosure, the second current **C2** may include currents flowing in opposite directions along the direction where the symmetric protrusion portion **2432** is extended. For example, with respect to the x-axis center of the symmetric protrusion portion **2432**, the second current **C2** may flow toward the negative x-axis direction, and the other second current **C2** may flow toward the positive x-axis direction.

(93) In an embodiment of the disclosure, when viewed from a position where the first current **C1** and the second current **C2** are incident on the protrusion **243**, the impedances of the symmetric protrusion portion **2432** in the negative x-axis direction and in the positive x-axis direction with respect to the x-axis center of the protrusion **243** may be formed to be the same.

(94) In an embodiment of the disclosure, the second currents **C2** flowing in opposite directions along the direction where the symmetric protrusion portion **2432** is extended may have vectors symmetric about the x-axis center of the symmetric protrusion portion **2432** and the same scalar. The second currents **C2** having symmetric vectors and the same scalar flow in opposite directions, so that the H-fields caused by the second currents **C2** may cancel each other out.

(95) FIG. **5** is a view illustrating a printed circuit board **240** including slits **242** according to an embodiment of the disclosure.

(96) In an embodiment of the disclosure, the printed circuit board **240** may be disposed to surround at least a portion of the electronic component **230**. For example, the printed circuit board **240** may be disposed to be spaced apart from the electronic component **230** in a first direction, a second direction, and/or a third direction of the electronic component **230**.

(97) Referring to FIG. **5**, a first region **241a**, a second region **241b**, and/or a third region **241c** of the printed circuit board **240** may be formed around the electronic component **230**. For example, the first region **241a** may be positioned in a negative x-axis direction with respect to the electronic component **230**, and the second region **241b** may be positioned in a positive x-axis direction with respect to the electronic component **230**. The third region **241c** may be positioned in a negative y-axis direction with respect to the electronic component **230**.

(98) Referring to FIG. **5**, a plurality of slits **242** may be formed in each of the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**. For example, three slits **242** may be disposed at intervals in the y-axis direction in each of the first region **241a** and the second region **241b** of the printed circuit board **240**. In the third region **241c** of the printed circuit board **240**, six slits **242** may be disposed at intervals in the x-axis direction.

(99) The number of slits **242** shown in FIG. 5 is illustrative, and the number of slits **242** formed on the printed circuit board **240** may be not limited thereto. For example, each of the first region **241a** and the second region **241b** of the printed circuit board **240** may include four or more slits **242**.

(100) In an embodiment of the disclosure, a protrusion **243** may be formed between plural slits **242**. For example, the protrusion **243** may be formed by a portion of the printed circuit board **240** that protrudes in one direction between two slits **242** while the slits **242** are formed in the printed circuit board **240**. In the first region **241a**, the protrusion **243** may be formed to extend in the positive x-axis direction, in the second region **241b**, the protrusion **243** may be formed to extend in the negative x-axis direction. In the third region **241c**, the protrusion **243** may be formed to extend in the positive y-axis direction.

(101) Referring to FIG. 5, a plurality of protrusions **243** may be formed in each of the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**. For example, two protrusions **243** may be disposed at intervals in the y-axis direction in each of the first region **241a** and the second region **241b** of the printed circuit board **240**. In the third region **241c** of the printed circuit board **240**, five protrusions **243** may be disposed at intervals in the x-axis direction. Each protrusion **243** may be positioned between two slits **242**.

(102) In an embodiment of the disclosure, slits **242** may be disposed in a spare space (e.g., empty space where no electronic component is arranged on the printed circuit board **240**) of a part of the printed circuit board **240**. In FIG. 5, since the third region **241c** of the printed circuit board **240** includes more spare space than the first region **241a** and the second region **241b**, the third region **241c** may include more slits **242** than the first region **241a** and the second region **241b**.

(103) FIG. 6 is a view illustrating a printed circuit board including a first slit **242** and a second slit **244** according to an embodiment of the disclosure.

(104) In an embodiment of the disclosure, the printed circuit board **240** may include a first slit **242** and/or a second slit **244**. For example, first slits **242** and second slits **244** may be formed in each of the first region **241a**, the second region **241b**, and the third region **241c** of the printed circuit board **240**.

(105) In an embodiment of the disclosure, the first slit **242** may refer to the slit **242** shown in FIG. 4. For example, the first slit **242** shown in FIG. 6 and the slit **242** shown in FIG. 4 may have the same shape. The first slit **242** may include a central slit portion **2421** and a symmetric slit portion **2422**.

(106) In an embodiment of the disclosure, the second slit **244** may be formed to extend in a direction away from the electronic component **230** and/or the cut region **241** on the printed circuit board **240**.

(107) In an embodiment of the disclosure, the first slit **242** and the second slit **244** may be alternately disposed along one direction of the printed circuit board **240**. For example, in each of the first region **241a** and the second region **241b**, the first slit **242** and the second slit **244** may be alternately arranged at intervals along the length direction (e.g., y-axis direction) of the printed circuit board **240**. In the third region **241c**, the first slit **242** and the second slit **244** may be alternately arranged at intervals along the width direction (e.g., x-axis direction) of the printed circuit board **240**.

(108) In an embodiment of the disclosure, a protrusion **245** may be formed between the first slit **242** and the second slit **244** that are alternately disposed along one direction of the printed circuit board **240**. The protrusion **245** may include an extension portion **2451** and a symmetric portion **2452**.

(109) Portion 'B' of FIG. 6 indicates an enlarged view of the protrusion **245** formed on the printed circuit board **240** according to an embodiment.

(110) In an embodiment of the disclosure, the extension portion **2451** of a protrusion **245** may be formed by a portion of the printed circuit board **240** that protrudes and is extended in a direction substantially perpendicular to the direction where the printed circuit board **240** is extended. For

example, in the third region **241c** of the printed circuit board **240**, the extension portion **2451** of the protrusion **245** may be formed by a portion of the printed circuit board **240** that is extended in a y-axis direction substantially perpendicular to the x-axis direction being the direction where the printed circuit board **240** is extended.

(111) In an embodiment of the disclosure, the extension portion **2451** of the protrusion **245** may protrude and extend in a direction where a portion of the printed circuit board **240** faces the electronic component **230**. For example, since the electronic component **230** is positioned in the positive y-axis direction with respect to the third region **241c** of the printed circuit board **240**, the extension portion **2451** may be extended in the positive y-axis direction.

(112) In an embodiment of the disclosure, the symmetric portion **2452** of the protrusion **245** may be extended from one end of the protrusion **245** in a direction perpendicular to the direction where the protrusion **245** is extended. For example, as indicated by portion 'B', the symmetric portion **2452** may be formed to extend from one end (e.g., end toward positive y-axis direction) of the protrusion **245** in a positive x-axis direction or a negative x-axis direction.

(113) In an embodiment of the disclosure, two protrusions **245** may be disposed with a second slit **244** interposed therebetween. Two protrusions **245** disposed with a second slit **244** interposed therebetween may be formed to be symmetric to each other with respect to the second slit **244**.

(114) In an embodiment of the disclosure, the symmetric portions **2452** of one protrusion **245** and another protrusion **245** spaced apart from each other with a second slit **244** interposed therebetween may be extended in opposite directions. For example, as indicated by portion 'B' of FIG. 6, the symmetric portion **2452** of one protrusion **245** may be extended toward the negative x-axis direction, and the symmetric portion **2452** of another protrusion **245** spaced apart from the one protrusion **245** in the positive x-axis direction may be extended toward the positive x-axis direction.

(115) In an embodiment of the disclosure, one end of a protrusion **245** may refer to the end of the protrusion **245** located closest to the cut region **241**. For example, in portion 'B', one end of the protrusion **245** may indicate the end of the protrusion **245** toward the positive y-axis direction.

(116) In an embodiment of the disclosure, a current may flow in the protrusion **245** of the printed circuit board **240**. For example, a first current **C1** may flow along a positive y-axis direction toward one end of the protrusion **245**.

(117) In an embodiment of the disclosure, the first current **C1** as a return current (**C-r**, see FIG. 8A) may be a current having a frequency lower than a preset reference value. The length of the protrusion **245** in the y-axis direction may be formed to be less than a preset length, which can prevent formation of an H-field due to the first current **C1**.

(118) Referring to FIG. 6, the second current **C2** may flow in a direction where the extension portion **2451** is extended (e.g., positive y-axis direction) and then flow in a direction (e.g., x-axis direction) perpendicular to the direction where the protrusion **245** is extended (e.g., positive y-axis direction).

(119) In an embodiment of the disclosure, the second current **C2** may flow along the direction where the symmetric portion **2452** of a protrusion **245** is extended. For example, the second current **C2** may flow toward a negative x-axis direction or a positive x-axis direction along the direction where the symmetric portion **2452** of the protrusion **245** is extended.

(120) In an embodiment of the disclosure, the second current **C2** may include currents flowing in opposite directions along the direction in which the symmetric portion **2452** is extended. For example, for one symmetric portion **2452** and another symmetric portion **2452** positioned with a second slit **244** interposed therebetween, the second current **C2** may flow toward the negative x-axis direction in the one symmetric portion **2452**, and the second current **C2** may flow toward the positive x-axis direction in the other symmetric portion **2452**.

(121) In an embodiment of the disclosure, the second currents **C2** flowing in opposite directions have symmetric vectors and the same scalar, so that the H-fields caused by the second currents **C2**

may cancel each other out.

(122) FIG. 7 is a view illustrating a protrusion **247** including a circular protrusion portion **2472** according to an embodiment of the disclosure.

(123) In an embodiment of the disclosure, a slit **242** formed in the printed circuit board **240** may include a concave portion **2423** and/or a concave extension portion **2424**. The concave portion **2423** may be a portion formed in a shape in which one side and the other side of the slit **242** are concave in a direction perpendicular to a direction where the slit **242** is extended.

(124) Referring to FIG. 7, the concave portion **2423** may have a shape concave toward the center of the slit **242** toward a negative x-axis direction or a positive x-axis direction.

(125) In an embodiment of the disclosure, the concave extension portion **2424** may be formed by a portion that is extended from one end of the concave portion **2423** toward the printed circuit board **240**.

(126) In an embodiment of the disclosure, a protrusion **247** including a central protrusion portion **2471** and/or the circular protrusion portion **2472** may be formed between two slits **242** including the concave portion **2423**.

(127) In an embodiment of the disclosure, the protrusion **247** may be formed by a portion of the printed circuit board **240** that is extended in one direction. For example, in FIG. 7, the protrusion **247** may be formed by a portion of the printed circuit board **240** that is extended toward a positive y-axis direction.

(128) In an embodiment of the disclosure, the central protrusion portion **2471** may be a portion of the printed circuit board **240** that protrudes and is extended in a direction perpendicular to the direction where the printed circuit board **240** is extended. For example, when the printed circuit board **240** is extended in the x-axis direction, the central protrusion portion **2471** may protrude and extend toward the y-axis direction perpendicular to the x-axis direction.

(129) In an embodiment of the disclosure, the protrusion **247** may include a circular protrusion portion **2472** formed in a circular shape at one end of the central protrusion portion **2471**. For example, the circular protrusion portion **2472** may be formed in a circular shape at an end of the central protrusion portion **2471** toward the positive y-axis direction.

(130) FIG. 7 shows a case where the protrusion **247** includes a circular protrusion portion **2472**, but this is illustrative, and a non-circular polygonal protrusion portion may be formed at one end of the central protrusion portion **2471**. For example, one end of the central protrusion portion **2471** may be formed to have a polygonal (e.g., pentagonal, hexagonal) shape that is symmetric with respect to a center line of the central protrusion portion **2471** in the x-axis direction.

(131) In an embodiment of the disclosure, a current may flow in the protrusion **247** of the printed circuit board **240**. For example, a first current **C1** may flow along the direction where the protrusion **247** is extended. For example, the first current **C1** may flow toward a positive y-axis direction.

(132) In an embodiment of the disclosure, the first current **C1** as a return current (C-r, see FIG. 8A) may be a current having a frequency lower than a preset reference value. The length of the protrusion **247** in the y-axis direction may be formed to be less than a preset length, which can prevent formation of an H-field due to the first current **C1**.

(133) Referring to FIG. 7, a second current **C2** may flow toward a direction where the protrusion **247** is extended (e.g., positive y-axis direction) and then flow in a direction (e.g., x-axis direction) perpendicular to the direction where the protrusion **247** is extended (e.g., positive y-axis direction).

(134) In an embodiment of the disclosure, the second current **C2** may flow toward one side and the other side of the circular protrusion portion **2472** of the protrusion **247**. For example, the second current **C2** may flow in the negative x-axis direction being one direction of the circular protrusion portion **2472** of the protrusion **245** or may flow toward the positive x-axis direction being the other direction of the circular protrusion portion **2472**.

(135) In an embodiment of the disclosure, the second current **C2** flowing in one direction of the

circular protrusion portion **2472** and the second current **C2** flowing in the other direction of the circular protrusion portion **2472** have symmetric vectors and the same scalar, so that the H-fields caused by the second currents **C2** may cancel each other out.

(136) FIGS. **8A**, **8B**, **8C** and **8D** are diagrams illustrating a return current **C-r** generated on a printed circuit board **240** and an equivalent circuit according to various embodiments of the disclosure.

(137) FIG. **8A** is a schematic diagram illustrating a return current **C-r** generated in the printed circuit board **240** when no slit **242** (see FIG. **4**) is formed around the cut region **241**.

(138) With reference to FIG. **8A**, a battery current **C-v** generated from the battery **210** (see FIG. **2**) may flow toward a power consuming component positioned on one surface of the printed circuit board **240** (e.g., power consuming component positioned inside the shield can **250** in FIG. **2**). For example, the connector (e.g., the battery connector **270** in FIG. **2**) connected to the battery **210** has a relatively high potential and the power consuming component has a relatively low potential, so that the battery current **C-v** may flow toward the power consuming component.

(139) A return current **C-r** may be generated due to the flow of the battery current **C-v**. The return current **C-r** may flow from the power consuming component toward a portion (e.g., ground portion) of the printed circuit board **240**. When no slit **242** (see FIG. **4**) is formed around the cut region **241**, the generated return current **C-r** may flow around the cut region **241**.

(140) FIG. **8B** is a diagram illustrating an H-field **H** generated due to a return current **C-r** when no slit **242** is formed around the cut region **241**.

(141) When no slit **242** (see FIG. **4**) is formed around the cut region **241**, a return current **C-r** flowing around the cut region **241** may be generated. For example, the return current **C-r** may flow along the outer circumference of the cut region **241** in a negative y-axis direction, positive x-axis direction, and positive y-axis direction in that order.

(142) When no slit **242** (see FIG. **4**) is formed around the cut region **241**, an H-field **H** may be formed in at least a part of the cut region **241** due to the return current **C-r**. An H-field **H** may be formed in a direction perpendicular to the flowing direction of the return current **C-r**.

(143) For example, with reference to FIG. **8B**, an H-field **H** may be formed along the circumference of a circle formed with respect to the flowing direction of the return current **C-r**. The H-field **H** due to the return current **C-r** may be formed toward a direction coming from one surface of the cut region **241** as indicated in FIG. **8B**.

(144) FIG. **8C** is a diagram illustrating a first circuit **810** being an equivalent circuit formed around the cut region **241** when no slit **242** (see FIG. **4**) is formed around the cut region **241**.

(145) When the printed circuit board **240** (see FIG. **4**) around the cut region **241** does not include a slit **242**, a first circuit **810** through which a current can flow may be formed around the cut region **241**. The first circuit **810** may include a voltage source **V**, but may not include a separate resistor therein. Since the first circuit **810** does not include a resistor, a current can flow in the first circuit **810** without hindrance.

(146) FIG. **8D** is a diagram illustrating a second circuit **820** being an equivalent circuit formed around the cut region **241** when slits **242** (see FIG. **4**) are formed around the cut region **241**.

(147) When the printed circuit board **240** (see FIG. **4**) around the cut region **241** includes slits **242** (see FIG. **4**), no current may flow around the cut region **241**. With reference to FIG. **4**, when the printed circuit board **240** includes slits **242**, protrusions **243** are formed on a part of the printed circuit board **240**, and currents flowing in opposite directions along symmetric protrusion portions **2432** of the protrusions **243** may be canceled out. As the current is canceled in the protrusions **243** and the flow of current is hindered, if represented as an equivalent circuit diagram, this may be represented as a second circuit **820** including a resistor **R** formed around the cut region **241**.

(148) In an embodiment of the disclosure, the second circuit **820** includes a voltage source **V** and a resistor **R**, and the resistor **R** has an infinite value, which can prevent a current from flowing inside the second circuit **820**. The resistor **R** in the second circuit **820** may indicate protrusions **243** (see

FIG. 4) of the printed circuit board **240** (see FIG. 4) formed through slits **242** (see FIG. 4). The resistor **R** in the second circuit **820** may mean that the protrusions **243** (see FIG. 4) impede the flow of current around the cut region **241**.

(149) FIGS. **9A** and **9B** are diagrams illustrating the intensity of the H-field (**H1**, **H2**) generated on the printed circuit board **240** according to various embodiments of the disclosure.

(150) FIG. **9A** is a diagram showing a simulation result for the intensity of an H-field **H1** generated on the printed circuit board **240** and therearound when the printed circuit board **240** does not include a slit **242** (see FIG. 4) around the cut region **241**.

(151) FIG. **9B** is a diagram showing a simulation result for the intensity of an H-field **H2** generated on the printed circuit board **240** and therearound when the printed circuit board **240** includes slits **242** (see FIG. 4) around the cut region **241**.

(152) Box lines **S** in FIGS. **9A** and **9B** indicate a portion where the cut region **241** is positioned in the printed circuit board **240**.

(153) In an embodiment of the disclosure, the H-field **H1** or **H2** may refer to a magnetic field generated by a current flowing in the printed circuit board **240**.

(154) Referring to FIGS. **9A** and **9B**, a portion where the intensity of the H-field **H1** or **H2** is strong may be displayed brighter on the drawing than a portion where the intensity of the H-field **H1** or **H2** is not strong. For example, a portion where the intensity of the H-field **H1** or **H2** is relatively strong may be displayed in relatively bright colors (e.g., green, yellow, red) on the drawing, a portion where the intensity of the H-field **H1** or **H2** is relatively weak may be displayed in relatively dark colors (e.g., blue) on the drawing.

(155) In an embodiment of the disclosure, when the printed circuit board **240** includes slits **242** (see FIG. 4) around the cut region **241**, compared to the case where the printed circuit board **240** does not include a slit **242** (see FIG. 4) around the cut region **241**, the intensity of the H-field **H1** or **H2** formed around the cut region **241** may be weak. When the printed circuit board **240** includes slits **242** (see FIG. 4) around the cut region **241**, as the intensity of the H-field **H1** or **H2** formed around the cut region **241** is relatively weak, the brightness around the cut region **241** in FIG. **9A** may be formed to be higher than the brightness around the cut region **241** in FIG. **9B**.

(156) In an embodiment of the disclosure, the printed circuit board **240** including slits **242** (see FIG. 4) around the cut region **241** makes the intensity of the H-field **H2** formed in the cut region **241** relatively weak, which can prevent or reduce malfunction of the electronic component **230** including a coil (see FIG. 4) disposed in the cut region **241**.

(157) An electronic device **200** according to an embodiment of the disclosure may include an electronic component **230** including a coil therein, and a printed circuit board **240** that is disposed to surround at least a part of the electronic component **230** and includes plural slits **242** disposed at preset intervals in at least one of regions facing the electronic component **230**, wherein each of the plural slits **242** may include a central slit portion **2421** extended along the center of the slit **242** and a symmetric slit portion **2422** extended to be symmetric with respect to the central slit portion **2421**.

(158) In an embodiment of the disclosure, the printed circuit board **240** may include a first region positioned to be spaced apart in a first direction of the electronic component **230** and facing the electronic component **230**, a second region positioned to be spaced apart from the electronic component **230** in a second direction opposite to the first direction and facing the electronic component **230**, and a third region positioned to be spaced apart from the electronic component **230** in a third direction perpendicular to the first and second directions and facing the electronic component **230**.

(159) In an embodiment of the disclosure, a plurality of slits **242** may be disposed at intervals in at least one region among the first, second, and third regions of the printed circuit board **240**.

(160) In an embodiment of the disclosure, a plurality of slits **242** may be disposed at intervals in each of the first region, the second region, and the third region of the printed circuit board **240**.

(161) In an embodiment of the disclosure, the printed circuit board **240** may include a cut region **241** formed by cutting off a portion of the printed circuit board **240** and where the electronic component **230** is disposed.

(162) In an embodiment of the disclosure, the printed circuit board **240** may include a protrusion **243** formed between plural slits **242** and extended in a direction where a portion of the printed circuit board **240** protrudes toward the electronic component **230**.

(163) In an embodiment of the disclosure, the central slit portion **2421** is formed to extend in a direction away from the electronic component **230**, and the symmetric slit portion **2422** may be extended from one end of the central slit portion **2421** in a direction perpendicular to a direction where the central slit portion **2421** is extended so as to be symmetric with respect to the central slit portion **2421**.

(164) In an embodiment of the disclosure, the printed circuit board **240** may include a protrusion **243** formed between plural slits **242** and extended in a direction where a portion of the printed circuit board **240** protrudes toward the electronic component **230**, and the protrusion **243** may include a central protrusion portion **2431** in which a portion of the printed circuit board **240** protrudes and extends in a direction perpendicular to the direction where the printed circuit board **240** is extended, and a symmetric protrusion portion **2432** formed to extend from one end of the central protrusion portion **2431** in a direction perpendicular to the direction where the central protrusion portion **2431** is extended in a form symmetric with respect to the central protrusion portion **2431**.

(165) In an embodiment of the disclosure, the slit **242** may include the concave portion **2423** formed in a shape in which one side and the other side are concave toward a direction perpendicular to the direction where the slit **242** is extended, and the concave extension portion **2424** connected to the concave portion **2423** and extended from one end of the concave portion **2423** toward the printed circuit board in a direction away from the electronic component **230**.

(166) In an embodiment of the disclosure, the printed circuit board **240** may include a protrusion **247** formed between plural slits **242** and extended in a direction where a portion of the printed circuit board **240** protrudes toward the electronic component **230**, and the protrusion **247** may include the central protrusion portion **2471** in which a portion of the printed circuit board **240** protrudes and extends in a direction perpendicular to the direction where the printed circuit board is extended, and the circular protrusion portion **2472** formed in a circular shape at one end of the central protrusion portion **2471**.

(167) In an embodiment of the disclosure, the slits **242** may include a first slit **242** including a central slit portion **2421** and a symmetric slit portion **2422**, and a second slit **244** extended in a direction away from the electronic component **230**, and the first slit **242** and the second slit **244** may be alternately disposed at intervals in at least a region of the printed circuit board **240**.

(168) In an embodiment of the disclosure, the printed circuit board **240** may include at least two protrusions **245** that are formed between plural slits **242** to protrude and extend in a direction where a portion of the printed circuit board **240** faces the electronic component **230**, a protrusion **245** may include an extension portion **2451** in which a portion of the printed circuit board **240** protrudes and extends in a direction toward the electronic component **230**, and a symmetric portion **2452** extended from one end of the extension portion **2451** in a direction perpendicular to the direction where the extension portion **2451** is extended, and the two protrusions **245** may be disposed with a second slit **244** interposed therebetween, and the two protrusions **245** disposed with a second slit **244** interposed therebetween may be formed to be symmetric to each other with respect to the second slit **244**.

(169) In an embodiment of the disclosure, the electronic component **230** may be a speaker or a receiver.

(170) In an embodiment of the disclosure, the electronic device **200** may further include a battery **210** disposed to be spaced apart from the printed circuit board **240** in a length direction of the

electronic device, the battery connector **270** disposed on the printed circuit board **240** and electrically connecting the printed circuit board **240** and the battery **210**, and a power consuming component disposed on the printed circuit board **240** and receiving power from the battery **210**, wherein the plural slits **242** may be formed in a region positioned relatively close to the power consuming component compared to other regions among the first region, the second region, and the third region.

(171) A printed circuit board **240** according to an embodiment of the disclosure may include a cut region **241** formed by cutting off a portion of the printed circuit board **240**, and plural slits **242** formed around the cut region **241**, wherein each of the plural slits **242** may include a central slit portion **2421** extended along the center of the slit **242** and a symmetric slit portion **2422** extended to be symmetric with respect to the central slit portion **2421**.

(172) The electronic device according to an embodiment of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic devices according to embodiments of the disclosure are not limited to those described above.

(173) It should be appreciated that embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(174) As used in connection with an embodiment of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, logic, logic block, part, or circuitry. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment of the disclosure, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(175) An embodiment as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium

and where the data is temporarily stored in the storage medium.

(176) According to an embodiment of the disclosure, a method according to an embodiment of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(177) According to an embodiment of the disclosure, each component (e.g., module or program) of the above-described components may include a singular or a plurality of entities, and some of the plurality of entities may be separately disposed in any other component. According to an embodiment of the disclosure, one or more components or operations among the above-described components may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., module or program) may be integrated into one component. In this case, the integrated component may perform one or more functions of each component of the plurality of components identically or similarly to those performed by the corresponding component among the plurality of components prior to the integration. According to an embodiment of the disclosure, operations performed by a module, program, or other component may be executed sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(178) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: an electronic component including a coil therein; and a printed circuit board that is disposed to surround at least a part of the electronic component and includes plural slits disposed at preset intervals in at least one of regions facing the electronic component, wherein each of the plural slits includes: a central slit portion extended along a center of the slit, and a symmetric slit portion extended to be symmetric with respect to the central slit portion.
2. The electronic device of claim 1, wherein the printed circuit board includes: a first region positioned to be spaced apart in a first direction of the electronic component and facing the electronic component; a second region positioned to be spaced apart from the electronic component in a second direction opposite to the first direction and facing the electronic component; and a third region positioned to be spaced apart from the electronic component in a third direction perpendicular to the first and second directions and facing the electronic component, and wherein the plural slits are disposed at intervals in at least one region among the first region, the second region, and the third region.
3. The electronic device of claim 2, wherein the plural slits are disposed at intervals in each of the first region, the second region, and the third region.
4. The electronic device of claim 2, further comprising: a battery disposed to be spaced apart from the printed circuit board in a length direction of the electronic device; a battery connector disposed on the printed circuit board and electrically connecting the printed circuit board and the battery; and a power consuming component disposed on the printed circuit board and receiving power from the battery, wherein the plural slits are formed in a region positioned relatively close to the power

consuming component compared to other regions among the first region, the second region, and the third region.

5. The electronic device of claim 1, wherein the printed circuit board includes a cut region formed by cutting off a portion of the printed circuit board and where the electronic component is disposed.

6. The electronic device of claim 1, wherein a portion of the printed circuit board includes a protrusion that is formed between plural slits and to protrude and extend in a direction toward the electronic component.

7. The electronic device of claim 1, wherein the central slit portion is formed to extend in a direction away from the electronic component, and wherein the symmetric slit portion is extended from one end of the central slit portion in a direction perpendicular to a direction where the central slit portion is extended so as to be symmetric with respect to the central slit portion.

8. The electronic device of claim 7, wherein a portion of the printed circuit board includes a protrusion that is formed between plural slits and to protrude and extend in a direction toward the electronic component, and wherein the protrusion includes: a central protrusion portion in which a portion of the printed circuit board protrudes and extends in a direction perpendicular to a direction where the printed circuit board is extended; and a symmetric protrusion portion formed to extend from one end of the central protrusion portion in a direction perpendicular to a direction where the central protrusion portion is extended in a form symmetric with respect to the central protrusion portion.

9. The electronic device of claim 1, wherein the slit includes: a concave portion formed in a shape in which one side and another side are concave toward a direction perpendicular to a direction where the slit is extended; and a concave extension portion connected to the concave portion and extended from one end of the concave portion toward the printed circuit board in a direction away from the electronic component.

10. The electronic device of claim 9, wherein a portion of the printed circuit board includes a protrusion that is formed between plural slits and to protrude and extend in a direction toward the electronic component, and wherein the protrusion includes: a central protrusion portion in which a portion of the printed circuit board protrudes and extends in a direction perpendicular to a direction where the printed circuit board is extended; and a circular protrusion portion formed in a circular shape at one end of the central protrusion portion.

11. The electronic device of claim 1, wherein the slits include: a first slit including a central slit portion and a symmetric slit portion; and a second slit formed to extend in a direction away from the electronic component, and wherein the first slit and the second slit are alternately disposed at intervals in at least a region of the printed circuit board.

12. The electronic device of claim 11, wherein a portion of the printed circuit board includes at least two protrusions that are formed between plural slits to protrude and extend in a direction toward the electronic component, wherein the protrusion includes: an extension portion in which a portion of the printed circuit board protrudes and extends in a direction toward the electronic component; and a symmetric portion extended from one end of the extension portion in a direction perpendicular to a direction where the extension portion is extended, wherein the at least two protrusions are disposed with a second slit interposed therebetween, and wherein the at least two protrusions disposed with a second slit interposed therebetween are formed to be symmetrical to each other with respect to the second slit.

13. The electronic device of claim 1, wherein the electronic component is a speaker or a receiver.

14. A printed circuit board comprising: a cut region formed by cutting off a portion of the printed circuit board; and a plurality of slits formed around the cut region, wherein each of the plural slits includes: a central slit portion extended along a center of the slit, and a symmetric slit portion extended to be symmetric with respect to the central slit portion.

15. The printed circuit board of claim 14, comprising: a first region positioned in a first direction with respect to the cut region; a second region positioned in a second direction opposite to the first

direction with respect to the cut region; and a third region positioned in a third direction perpendicular to the first and second directions with respect to the cut region, wherein the plural slits are disposed at intervals in at least one region among the first region, the second region, and the third region.

16. The printed circuit board of claim 15, wherein the plural slits are disposed at intervals in each of the first region, the second region, and the third region.

17. The printed circuit board of claim 14, wherein the central slit portion is formed to extend in a direction away from the cut region, and wherein the symmetric slit portion is extended from one end of the central slit portion in a direction perpendicular to a direction where the central slit portion is extended so as to be symmetric with respect to the central slit portion.

18. The printed circuit board of claim 17, wherein a portion of the printed circuit board includes a protrusion that is formed between plural slits and to protrude and extend in a direction toward the cut region, and wherein the protrusion includes: a central protrusion portion formed by a portion of the printed circuit board that protrudes and extends in a direction perpendicular to a direction where the printed circuit board is extended; and a symmetric protrusion portion formed to extend from one end of the central protrusion portion in a direction perpendicular to a direction where the central protrusion portion is extended in a form symmetric with respect to the central protrusion portion.

19. The printed circuit board of claim 14, wherein the slit includes: a concave portion formed in a shape in which one side and another side are concave toward a direction perpendicular to a direction where the slit is extended; and a concave extension portion connected to the concave portion and extended from one end of the concave portion toward the printed circuit board in a direction away from the cut region.

20. The printed circuit board of claim 14, wherein the slits include: a first slit including a central slit portion and a symmetric slit portion, and a second slit formed to extend in a direction away from the cut region, and wherein the first slit and the second slit are alternately disposed at intervals.
